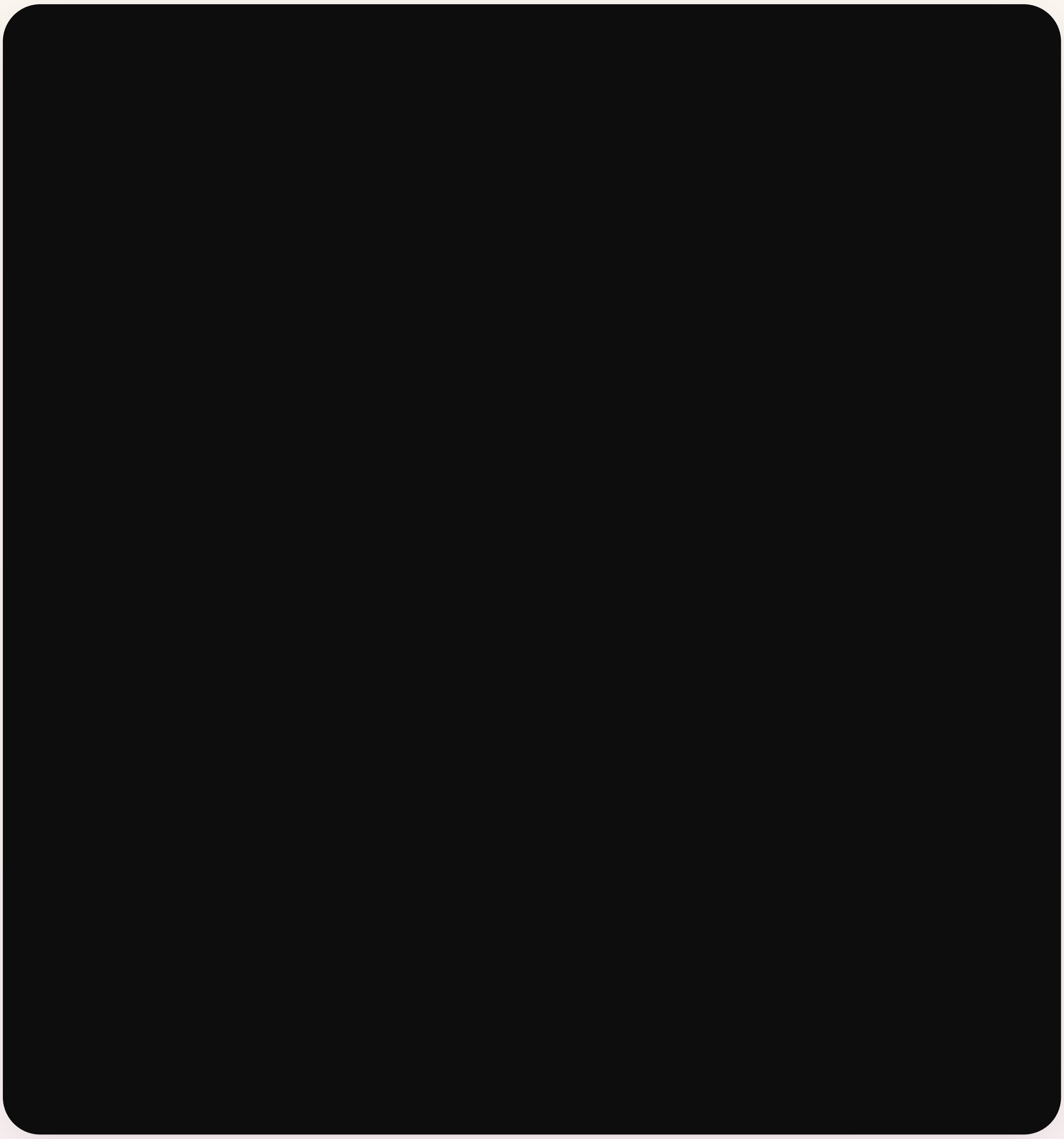
varun's Bedtime Adventure

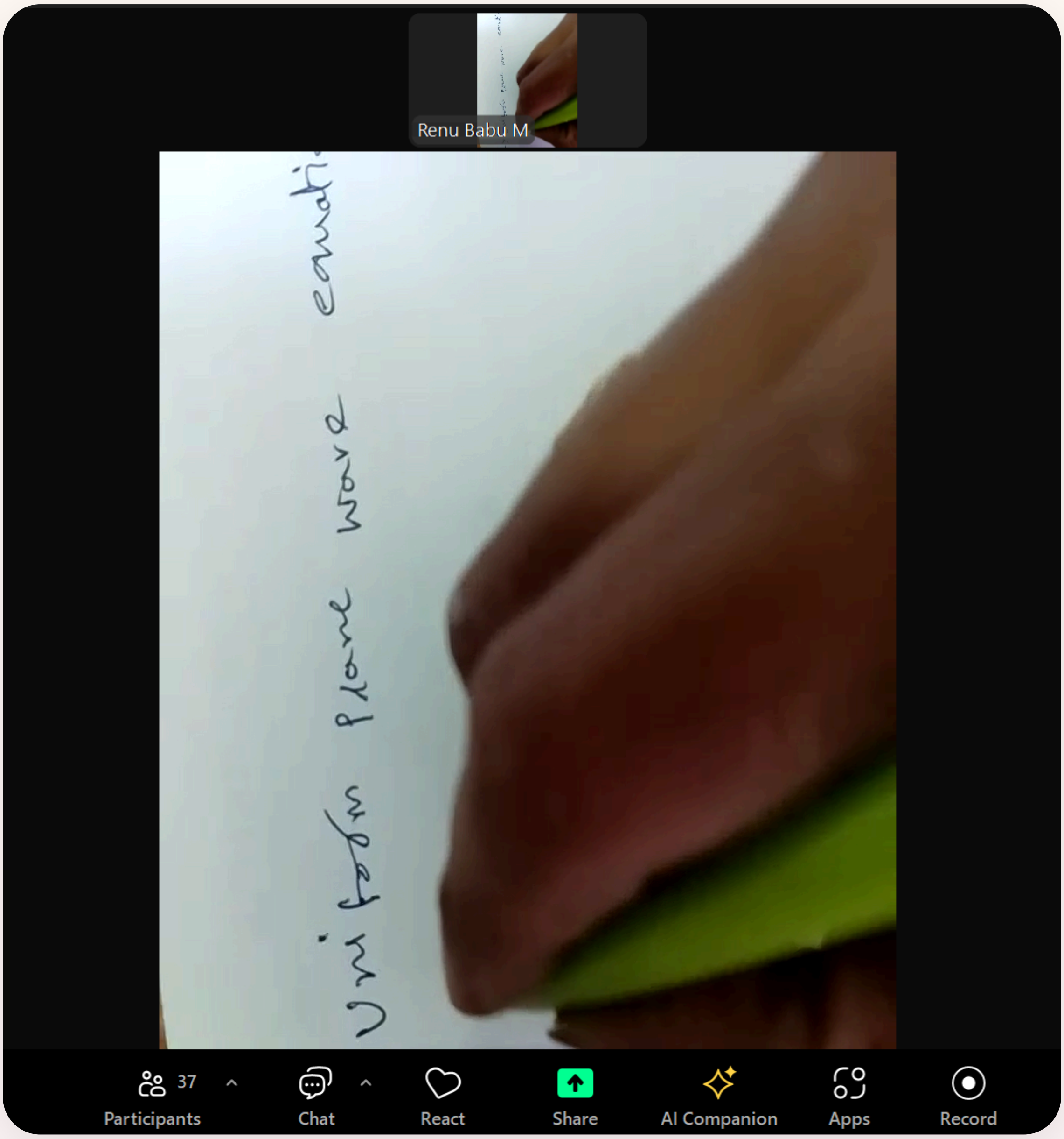
A Magical Journey to Dreamland

Bath Time



Splish, splash! Little varun loves bath time. The warm water feels so nice, and the bubbles dance and pop. varun giggles as tiny rubber ducks float by. Clean and cozy, it's time for the next adventure!

Pajama Time



Now it's time for the softest, coziest pajamas! varun stretches little arms up high as sowmya helps with the buttons. The pajamas have the sweetest pattern, perfect for a night full of wonderful dreams.

Family Cuddle

7.5.1 Finite State Machine—Definitions

Consider the state diagram of a finite state machine shown in Figure 7.9. It is a five-state machine with one input variable and one output variable.

$$S = \{A, B, C, D, E\} \quad I = \{0, 1\} \quad O = \{0, 1\}$$

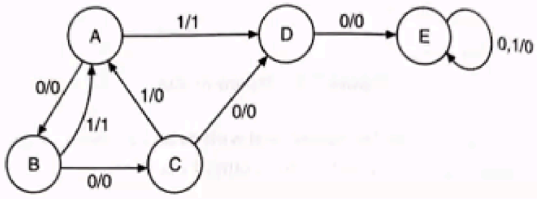


Figure 7. 9 State diagram.

Successor: Looking at the state diagram in Figure 7.9 we can say that, when present state is A and input is 1, the next state is D. In other words, this condition is specified as D is the 1-successor of A. Similarly, we can say that A is the 1 successor of B and C, D is the 11 successor of B and C, C is the 00 successor of A, D is the 000 successor of A, E is the 10 successor of A or 0000 successor of A and so on. In general, we can say that, if an input sequence X takes a machine from state S_i to S_j, then S_j is said to be the X successor of S_i.

Terminal state: Looking at the state diagram of Figure 7.9, we observe that no such input sequence exists which can take the sequential machine out of state E and thus state E is said to be a terminal state.

In general, we can say that a state is a terminal state when there are no outgoing arcs which start from it and terminate in other states.

Strongly-connected machine: In sequential machines many times certain subsets of states may not be reachable from other subsets of states, even if the machine does not contain any terminal state. However, if for every pair of states S_i, S_j of a sequential machine there exists an input sequence which takes the machine M from S_i to S_j, then the sequential machine is said to be strongly connected.

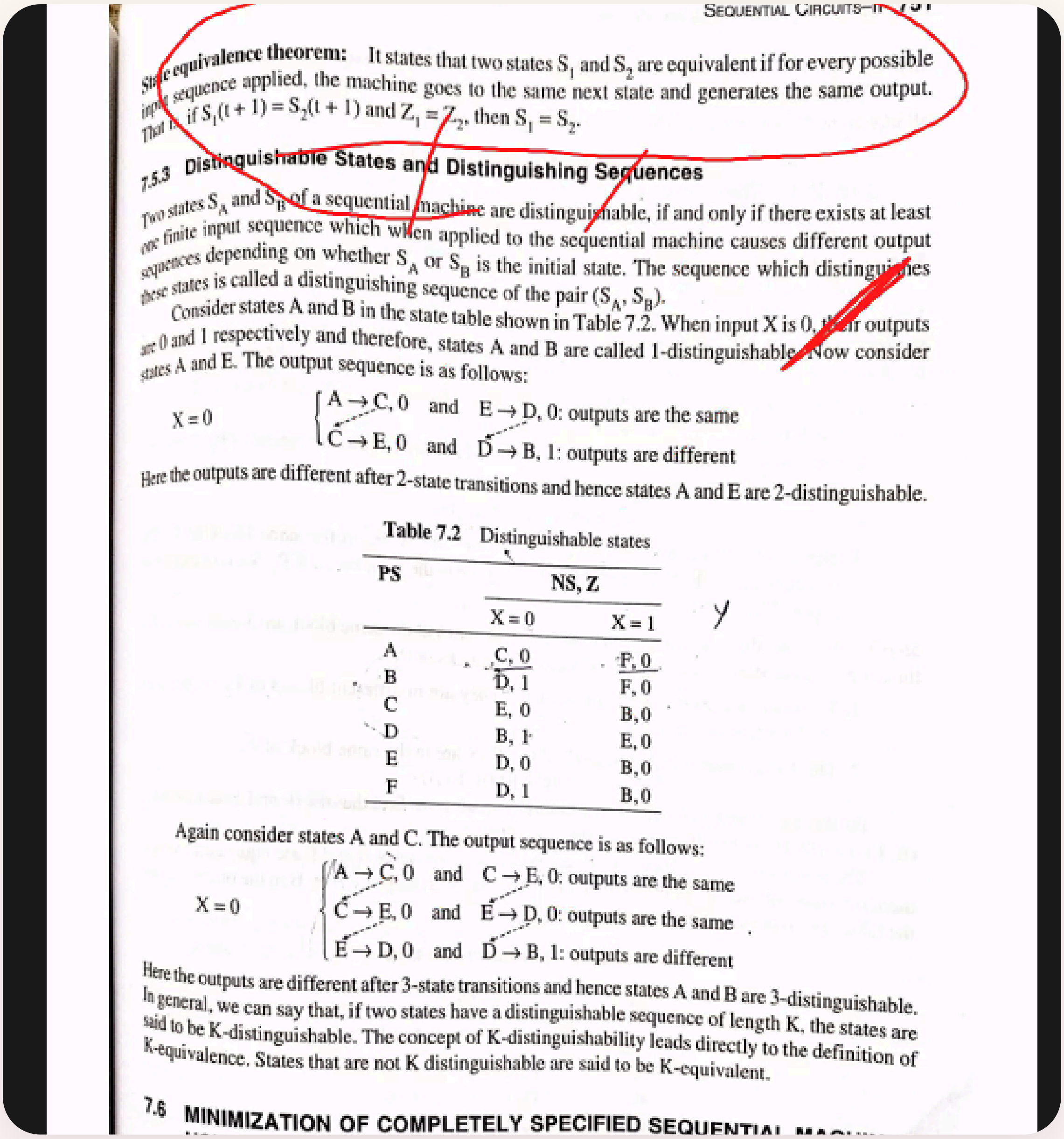
7.5.2 State Equivalence and Machine Minimization

In realizing the logic diagram from a state table or state diagram many times we come across redundant states. Redundant states are states whose functions can be accomplished by other states. The elimination of redundant states reduces the total number of states of the machine which in turn results in reduction of the number of flip-flops and logic gates, reducing the cost of the final circuit.

Two states are said to be equivalent, if every possible set of inputs applied to the machine while in this state generate exactly the same output and take the machine exactly to the same next state. When two states are equivalent, one of them can be removed without altering the input-output relationship. Here we will discuss what is meant by state equivalence and how to find equivalent states for machine minimization.

The best part of the evening is cuddle time! varun snuggles close with sowmya, feeling warm and safe. Little arms wrap around for the biggest hug. Love fills the room like a cozy blanket.

Story Time



State equivalence theorem: It states that two states S_1 and S_2 are equivalent if for every possible input sequence applied, the machine goes to the same next state and generates the same output. That is, if $S_1(t + 1) = S_2(t + 1)$ and $Z_1 = Z_2$, then $S_1 = S_2$.

7.5.3 Distinguishable States and Distinguishing Sequences

Two states S_A and S_B of a sequential machine are distinguishable, if and only if there exists at least one finite input sequence which when applied to the sequential machine causes different output sequences depending on whether S_A or S_B is the initial state. The sequence which distinguishes these states is called a distinguishing sequence of the pair (S_A, S_B).

Consider states A and B in the state table shown in Table 7.2. When input X is 0, their outputs are 0 and 1 respectively and therefore, states A and B are called 1-distinguishable. Now consider states A and E. The output sequence is as follows:

$$X = 0 \quad \begin{cases} A \rightarrow C, 0 & \text{and} \quad E \rightarrow D, 0\text{: outputs are the same} \\ C \rightarrow E, 0 & \text{and} \quad D \rightarrow B, 1\text{: outputs are different} \end{cases}$$

Here the outputs are different after 2-state transitions and hence states A and E are 2-distinguishable.

Table 7.2 Distinguishable states

PS	NS, Z	
	X = 0	X = 1
A	C, 0	F, 0
B	D, 1	F, 0
C	E, 0	B, 0
D	B, 1	E, 0
E	D, 0	B, 0
F	D, 1	B, 0

Again consider states A and C. The output sequence is as follows:

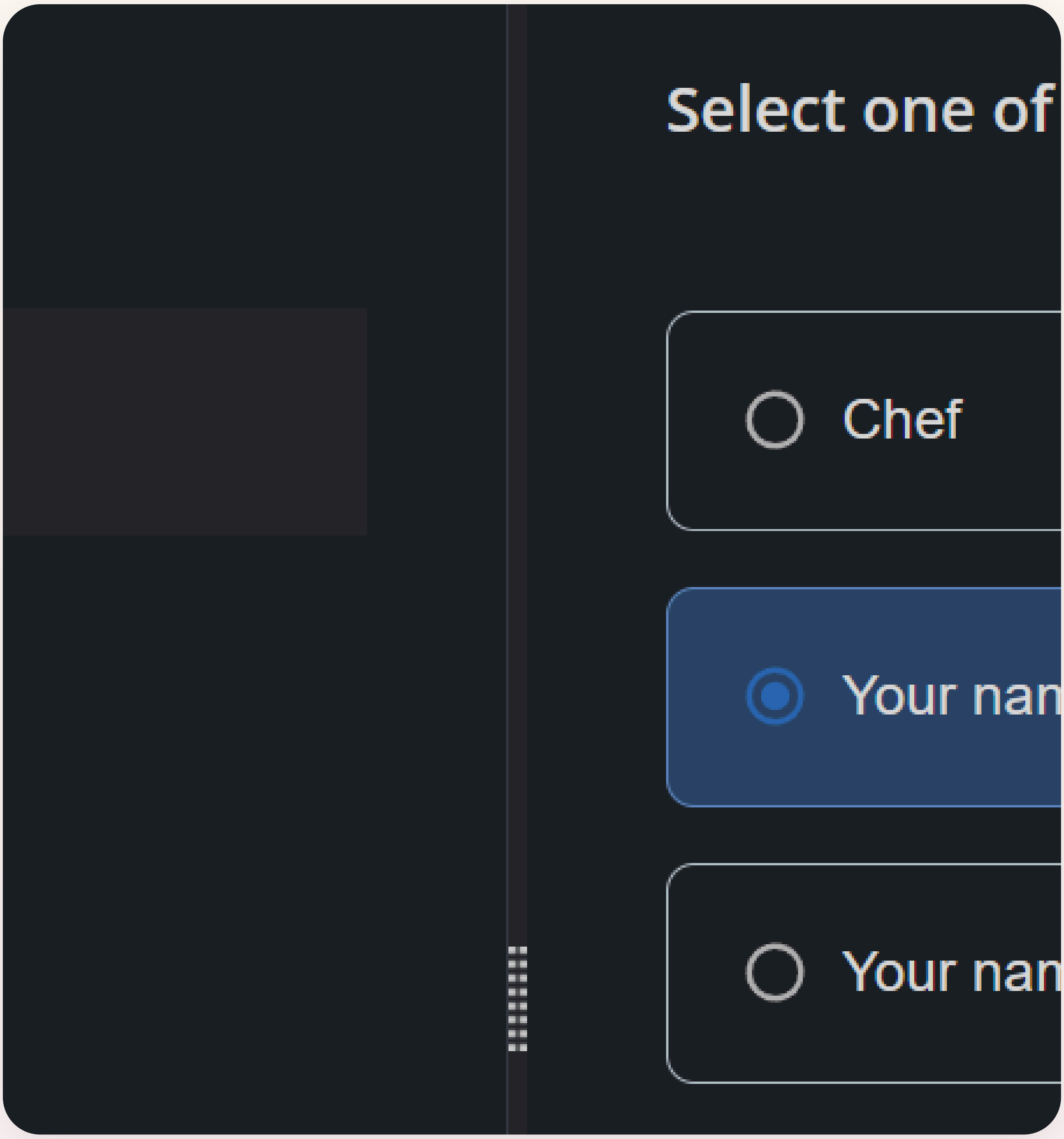
$$X = 0 \quad \begin{cases} A \rightarrow C, 0 & \text{and} \quad C \rightarrow E, 0\text{: outputs are the same} \\ C \rightarrow E, 0 & \text{and} \quad E \rightarrow D, 0\text{: outputs are the same} \\ E \rightarrow D, 0 & \text{and} \quad D \rightarrow B, 1\text{: outputs are different} \end{cases}$$

Here the outputs are different after 3-state transitions and hence states A and B are 3-distinguishable. In general, we can say that, if two states have a distinguishable sequence of length K, the states are said to be K-distinguishable. The concept of K-distinguishability leads directly to the definition of K-equivalence. States that are not K distinguishable are said to be K-equivalent.

7.6 MINIMIZATION OF COMPLETELY SPECIFIED SEQUENTIAL MACHINES

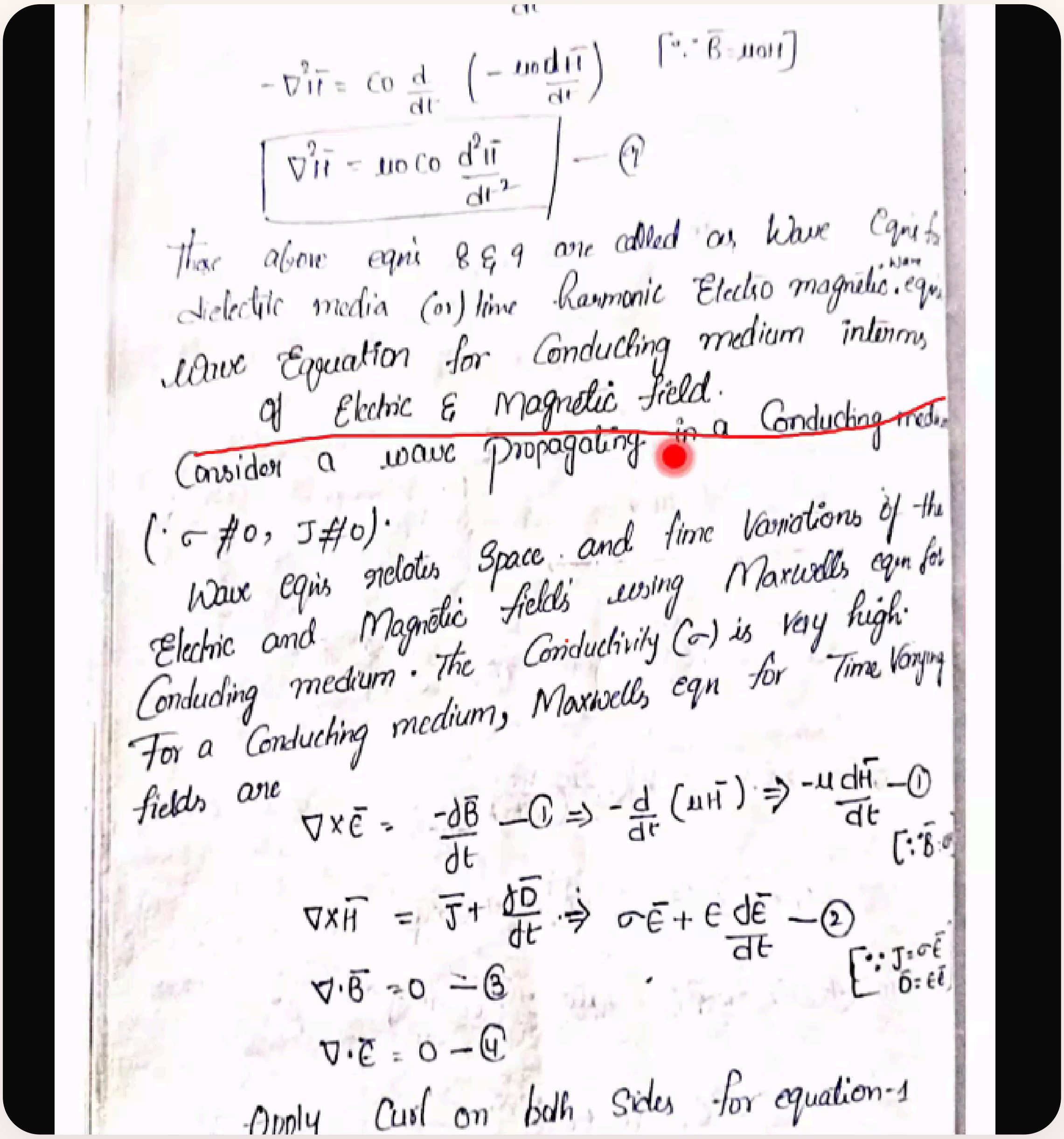
With a favorite book in hand, varun listens to magical tales of faraway lands. Each page brings new wonders and adventures. varun's eyes grow wide with amazement at every turn of the page.

Lullaby Time



A gentle song fills the air as sowmya hums a sweet lullaby. The melody dances through the room like fireflies in summer. varun's eyes begin to feel heavy, filled with peaceful dreams waiting to begin.

Goodnight Kiss



$$-\nabla^2 \bar{H} = \epsilon_0 \frac{d}{dt}\left(-\mu_0 \frac{d\bar{H}}{dt}\right) \quad [\because \bar{B} = \mu_0 \bar{H}]$$

$$\boxed{\nabla^2 \bar{H} = \mu_0 \epsilon_0 \frac{d^2 \bar{H}}{dt^2}} \text{ — } (9)$$

These above eqns 8 & 9 are called as Wave Eqns for dielectric media (or) time harmonic Electro magnetic wave eqns.

Wave Equation for Conducting medium interms of Electric & Magnetic field.

Consider a wave propagating in a Conducting medium ($\because \sigma \neq 0, J \neq 0$).

Wave eqns relates Space and time Variations of the Electric and Magnetic fields using Maxwells eqn for Conducting medium. The Conductivity (σ) is very high. For a Conducting medium, Maxwells eqn for Time Varying fields are

$$\nabla \times \bar{E} = -\frac{d\bar{B}}{dt} \text{ — } (1) \Rightarrow -\frac{d}{dt}(\mu \bar{H}) \Rightarrow -\mu \frac{d\bar{H}}{dt} \text{ — } (1) \quad [\because \bar{B} = \mu \bar{H}]$$

$$\nabla \times \bar{H} = \bar{J} + \frac{d\bar{D}}{dt} \Rightarrow \sigma \bar{E} + \epsilon \frac{d\bar{E}}{dt} \text{ — } (2) \quad [\because \bar{J} = \sigma \bar{E}, \ \bar{D} = \epsilon \bar{E}]$$

$$\nabla \cdot \bar{B} = 0 \text{ — } (3)$$

$$\nabla \cdot \bar{E} = 0 \text{ — } (4)$$

Apply Curl on both Sides for equation-1

One sweet kiss on varun's forehead, then another on each cheek. 'Goodnight, little one,' whispers sowmya softly. The nightlight glows warmly, keeping watch through the night.

Sweet Dreams

And now, dear varun drifts off to dreamland, where unicorns play and stars sing lullabies. Tomorrow brings new adventures, but for now, sleep tight, little one. Sweet dreams until morning light.